KVR

NEET — DPT7

21 August 2026

Instructions

1. This paper contains 20 questions. All questions are compulsory.
2. Each question has four options (A, B, C, D) with exactly one correct answer.
3. Mark your answers clearly. There is no negative marking unless announced.

1. Which cell organelle is known as the powerhouse of the cell?

(A) Nucleus
(B) Mitochondria
(C) Ribosome
(D) Golgi apparatus

2. A codon consists of how many nucleotides?

(A) 1
(B) 2
(C) 3
(D) 4

3. Assertion: DNA is a self-replicating molecule. Reason: DNA has complementary base pairing.

(A) Both A and R true, R correct explanation
(B) Both A and R true, R not correct explanation
(C) A true, R false
(D) A false, R true

4. The pacemaker of the human heart is:

(A) AV node
(B) SA node
(C) Bundle of His
(D) Purkinje fibres

5. Insulin is secreted by:

(A) Alpha cells of pancreas
(B) Beta cells of pancreas
(C) Adrenal gland
(D) Thyroid gland

6. Transpiration in plants mainly occurs through:

(A) Roots
(B) Stomata
(C) Stem
(D) Flowers

7. The site of protein synthesis in a cell is:

(A) Nucleus
(B) Ribosome
(C) Lysosome
(D) Mitochondria

8. Which pigment absorbs light most efficiently in photosynthesis?

(A) Chlorophyll a
(B) Chlorophyll b
(C) Carotene
(D) Xanthophyll

9. The bond angle in a water molecule is approximately:

(A) 90 degrees
(B) 104.5 degrees
(C) 109.5 degrees
(D) 120 degrees

10. Which of the following is an example of a carbohydrate?

(A) Glycine
(B) Glucose
(C) Glycerol
(D) Guanine

11. The pH of a 0.001 M HCl solution is:

(A) 1
(B) 2
(C) 3
(D) 4

12. Assertion: Coordination compounds show optical isomerism. Reason: They have non-superimposable mirror images.

(A) Both A and R true, R correct explanation
(B) Both A and R true, R not correct explanation
(C) A true, R false
(D) A false, R true

13. A pi bond is formed by the overlap of:

(A) Two s orbitals
(B) One s and one p orbital
(C) Two p orbitals sideways
(D) Two p orbitals head-on

14. The number of moles in 44 g of CO2 is:

(A) 1
(B) 2
(C) 0.5
(D) 4

15. A machine has 60% efficiency. If input energy is 500 J, the useful output energy is:

(A) 200 J
(B) 250 J
(C) 300 J
(D) 400 J

16. A projectile is fired at angle 45 degrees with initial velocity u. Its horizontal range is:

(A) u^2/g
(B) u^2/2g
(C) u^2/4g
(D) 2u^2/g

17. The coefficient of friction depends primarily on:

(A) Area of contact
(B) Nature of surfaces in contact
(C) Weight of the body
(D) Velocity of the body

18. A convex lens of focal length 20 cm forms a real image at 40 cm. The object distance is:

(A) 40 cm
(B) 20 cm
(C) 30 cm
(D) 60 cm

19. Work done by a centripetal force on a body moving in a circle is:

(A) Positive
(B) Negative
(C) Zero
(D) Depends on the radius

20. Which of the following has the dimensions of energy?

(A) [MLT^-1]
(B) [ML^2T^-2]
(C) [MLT^-2]
(D) [ML^-1T^-2]